

I am an Electrical & Computer Engineering student at UT Austin who wakes up every day driven by one question: how can robotics meaningfully improve people’s lives? Curiosity and creativity pull me toward building systems that help robots learn, adapt, and act in ways that empower people at home, at work, and at scale. Growing up in Peru and leading diverse teams gave me a unique understanding of people and possibility, teaching me to listen with empathy, set bold goals, and push through every obstacle until challenge becomes opportunity.

SKILLS

- AI & Robot Learning: ML, VLA Models, Diffusion Policies, Imitation Learning
 - Robotics: Computer Vision, Embedded Systems, Simulation, Teleoperation
- Robotics Frameworks: ROS, MuJoCo, RoboSuite, ManiSkill-HAB
 - Programming: Python, PyTorch, C, SQL, Assembly, OpenCV

EDUCATION

THE UNIVERSITY OF TEXAS AT AUSTIN, Austin, TX

B.S. in Electrical and Computer Engineering | Robotics Minor | GPA: 3.7 | University Honors: Fall 24, Fall 25, Spring 26

MARKHAM COLLEGE, Lima, Peru – International Baccalaureate Diploma

EXPERIENCE

GENERAL MOTORS – Autonomous Robotics Center, Detroit, MI

Manufacturing Engineer Intern | Enabled dexterous manipulation via diffusion and VLA policies, furthering an autonomous vehicle assembly line

- Contributed to dual-arm policies, mapping egocentric RGB-D vision and joint-state history to precise 6-DoF end-effector movements.
- Helped design multi-modal data pipeline (EgoDex dataset, human demos, robot teleoperation) to pretrain and fine-tune manipulation models.
- Recorded 200+ egocentric demos via a custom helmet-mounted depth-camera rig, and 20+ robot teleoperation demos using a force-feedback GELLO interface to control dual UR5e arms, supplying task-specific and high-precision training data for part-extraction policy fine-tuning.
- Applied computer vision (MediaPipe + CoTracker) to extract 3D hand-contact trajectories, converting raw video into structured training data.

YAPE, Lima, Peru

Data Scientist Intern | Empowered data-driven decision-making, expanded financial inclusion in underserved markets

- Automated high-volume transaction processing in Python, driving strategic decisions at Peru’s leading mobile-payment platform (20M+ users).
- Achieved \$500K in yearly savings and accelerated user and feature growth by building SQL dashboards for product and sales teams.
- Accelerated bill payment user adoption among merchants and enabled targeted outreach by leading high-impact data-driven initiatives.

RESEARCH & PROJECTS

Learning Agents Research Group – Undergraduate Researcher at UT Austin | Design scalable long-term memory for robotics

Developed custom multi-step, long-horizon robotic manipulation benchmarks in household environments using ManiSkill-HAB benchmark.

- Evaluated and deployed pre-trained RL policies for task execution involving navigation, object interaction, and articulated manipulation.
- Designed a memory-centric system to encode, store, and leverage past task-relevant information for long-horizon, multi-stage manipulation.

Electrical Team – ASCEND at UT Austin | Engineer embedded software and hardware for autonomous flight

- Developing an autonomy-enabled ATV and paramotor system to navigate, interact, and perform complex tasks in harsh environments.
- Designed and soldered battery harness, and wrote control code to drive motors for parafoil brake actuation, rebuilding dual-actuator system.
- Built dual load cell sensing system with amplifier/ADC circuits for live brake tension feedback, closed-loop control, and collapse detection.

Research Initiative on Robot Learning at UT Austin | Used imitation/reinforcement learning, improved robot task success

- Trained a Sawyer robot to roll a ball to a target using imitation and reinforcement learning with Robosuite, MuJoCo and PyTorch.
- Improved robot task success by automating computer-vision data collection for robotic manipulation, using LLM to clarify vague instructions.

Connect 4 – Intro to Embedded Systems at UT Austin | Engineered Connect 4 game with real-time feedback

- Programmed a TI microcontroller in C and Assembly. Interfaced toggle switches, display, speaker, and potentiometer to synchronize SW/HW.

LEADERSHIP

IEEE ROBOTICS AND AUTOMATION SOCIETY (RAS) at UT Austin, **Active Member**

SOCIETY OF HISPANIC PROFESSIONAL ENGINEERS (SHPE) at UT Austin, **Active Member**

YNG (YPO NEXT GENERATION) GLOBAL NETWORK (of 30,000+ members), Austin, TX and Lima, Peru

- **Engagement Chair**, YNG Austin - Oversee execution of local activities to increase chapter engagement

CLOUDS FOR CHANGE at Markham College, **Co-Founder & Head of Logistics**

- Led the design and execution of a fog-water harvesting project to provide sustainable water sources to underserved communities. Managed all logistics and fundraising efforts, securing over \$45K in donations to install 30 fog catchers supplying clean water to hundreds of people.

COCHRANE HOUSE at Markham College, **House Captain**

- Elected captain of 400+ member House. Won "Markham on Stage", school's most important inter-house contest for the first time in 5 years.

PERSONAL

Dual citizenship: United States and Peru. Fluent in English and Spanish. Enjoy playing guitar, soccer, working out, reading and cooking.